High risk geographic mobility patterns among young persons who inject drugs

Introduction

In the United States (U.S.), alarming increases in injection drug use (IDU)[1,2] and hepatitis C virus (HCV) infection incidence[3-7] are occurring among a new generation of young persons who inject drugs (PWID) from non-urban communities[4,8-14]. IDU is a well-established risk factor for HIV and HCV is and the primary mode of HCV transmission in developed countries[15]. Until recently, young PWID from non-urban areas, a population predominantly composed of non-Hispanic (NH) whites who engage in high levels of risky injection practices[4,5,9,10,12,16], generally exhibited low to modest incidence and prevalence of HIV and HCV infection[17-19]. This disjunction between behavior and HIV/HCV infection status was likely due to social and environmental factors that minimized the likelihood of having an HCV-infected partner[17]. Recent trends suggest that these protections are eroding. A high incidence of HCV infection is now common among non-urban PWID, with the highest rates observed among those aged 20–29 years[3-5]. Moreover, high levels of risky injection and sexual behaviors[20,21] and sexually-transmitted infections (STIs) [22] are observed among PWID, who report more sexual partners, greater likelihood of trading sex for money or drugs, and less consistent condom use than their non-drug using counterparts[23]. Understanding individual, social and geographic factors that promote high-risk partner contact is imperative for developing targeted HIV/HCV interventions for the emerging population of young PWID.

Geographic mobility has important implications for HIV/HCV risk and prevention among PWID. Previous studies have examined various aspects of mobility, including homelessness, residential transience (i.e., having multiple residences over a short period) and distances (e.g.,

travel) on HIV/HCV risk. Studies have consistently linked residential instability with high risk behaviors, including syringe sharing and sex exchange[24-27], and HIV infection[28,29]. A plausible mechanism for this may be via disrupting control of one's physical environment, fostering use of high risk strategies to acquire resources including injecting in public spaces, syringe sharing, and exchanging sex for drugs or money[25]. Additionally, residential transience may affect HIV/HCV risk by altering social networks (e.g., size, composition and member characteristics). Nonetheless, little is known about the impact of directional mobility, i.e., residential transience between neighborhoods with varying HIV/HCV prevalence on HIV/HCV risk. No known study has examined these mobility patterns and their association with HIV/HCV risk among the emerging population of young, predominantly non-urban PWID. Understanding these patterns is important because of the potential for disease transmission due to bridging between populations of high and low prevalence. Geographic mobility also presents challenges to the effective provision of interventions, and thus, understanding patterns of residential transience and associated risk are important for effective intervention development and targeting for this population.

Our current study sought to characterize mobility patterns, including residential transience and direction of movement (e.g., between urban and suburban areas and in relation to local drug market areas) and their association with sociodemographic, behavioral, and network characteristics of young PWID from both urban and surrounding Chicago suburban areas.

Methods

Subjects and Recruitment

Data were collected as part of a cross-sectional personal (egocentric) network study of young PWID and their injection, social support, and sexual networks. Detailed description of sample recruitment, study design, and data collection methods is previously described[27]. Briefly, participants were recruited via flyers posted at four standalone field sites in Chicago, Illinois, U.S. located near major heroin and cocaine markets that attract both urban and suburban drug users as well at venues that provide services to PWID in these areas. All individuals responding to posted flyers were directed to on-site research staff for eligibility screening during regular hours of operation between September 2012 and June 2013. Most participants were registered members of a large Chicago syringe exchange program (SEP) that operated out of four field sites and a mobile unit. To be eligible for the study, individuals had to be between the ages of 18 and 30 and have injected drugs at least once in the past 30 days. Participants were interviewed by trained research staff to collect all data. Participants self-reported demographic information, HIV and HCV status, drug and sexual behaviors for the past 6 months, and geographic data on the location and characteristics of places where they resided, purchased drugs, injected drugs, and met sex partners in the past year. Participants also provided similar data on each of their core injection network members, defined as up to the 10 individuals they injected most often with in the past 6 months. All participants provided written informed consent. The study was approved by the Institutional Review Board of the University of Illinois at Chicago.

Measures

In addition to sociodemographic characteristics, injection and sexual behaviors, participants provided the geographic locations (cross-streets, town, state) of all places where they resided, purchased drugs, injected drugs, and met sex partners in the past year. Participants also reported on recent homelessness, defined as living on the street or in a shelter at any time during the past

6 months. Using a substantive body of research among Chicago PWID in the past 30 years[17,30-36] along with published data on heroin and cocaine police arrest locations for Illinois[37], we suggest that there are three primary drug market areas, all of which are located with urban Chicago areas. Using conservative street boundaries, we created these three urban drug market areas aand named West, South, Southeast drug markets to correspond to regions of the city where they are located. Movement distance, defined as the average distance between residences for those with multiple residences in the past year, was calculated. Our main outcome of interest was patterns of geographic mobility, which we assessed in three ways. First, residential transience was defined as having more than one residence in the past year, regardless of location, as a binary outcome (yes/no). Second, using a 3-category indicator variable termed residential movement, we assessed whether PWID were remaining within the same neighborhood boundaries (urban/suburban Chicago), regardless of number of residences, or moving between neighborhoods over time. Participants were grouped into (i) urban, i.e., having $\geq$ 1 residence(s) exclusively within the Chicago city limits (referent category), (ii) suburban, i.e., having $\geq$ 1 residence(s) exclusively in the suburban areas surrounding Chicago, or (iii) crossover, i.e., having $\geq$1 residence(s) in urban *and* $\geq$1 residence(s) in the suburban areas in the past year. Third, among participants who reported transience (n=96), we assigned a variable termed bridging transience that compared localized transience (i.e., having multiple residences in the same neighborhood) vs. crossover transience (i.e., category iii in residential movement).

Participants also provided data by proxy on their core injection and sexual network members, including sociodemographic characteristics, residence status, mobility, homelessness, injection behaviors (e.g., syringe sharing), sexual risk behaviors (e.g., exchange sex, multiple sex partners), and HIV and HCV infection status. Summary network measures (e.g., size, proportion

reporting syringe sharing) were calculated by summarizing across all injection partnerships reported by the ego.

Statistical analysis

Sociodemographics, injection practices, and network characteristics were compared by residential movement using Pearson chi-square tests for categorical variables and Kruskal-Wallis tests for continuous variables. Multivariable logistic regression was used to identify factors associated with residential transience and crossover transience, while multinomial logistic regression was used to examine factors associated with residential movement. For all models, variables with $p<0.2$ in univariable analysis and those that were considered conceptually important were entered in multivariable models and non-significant variables were removed one by one in an iterative process in order to build a parsimonious final model. Gender, age, and race/ethnicity were included in all multivariable models regardless of statistical significance for consistency with prior literature and to control for residual confounding. Inferential statistical analyses were conducted using STATA/SE version 14 for Windows (STATA Corp, College Station, TX).

We also performed geographic analyses using Python (Python Software Foundation, Beaverton, OR) and R version 3.2 using the spatstat package. Geographic locations were geocoded for analysis using the Google Geocoding API with GeoPy (Google, Mountainview, CA). Locations were categorized according to location in urban or suburban areas and according to whether they were within the largest of the drug market area (West side) (yes/no). Maps were created to visualize relationships among residence, drug market boundaries, injection, and sex partner meeting locations (see Fig. 1). Logistic regression models with generalized estimating

equations (GEE) were used to compare injection and sex partner meeting locations (urban/suburban and within West side drug market) by residential movement status.

Results

Participant characteristics by residential movement status are shown in Table 1. Compared to urban and suburban, crossover participants were most likely to report, in the past 6 months, homelessness (81%), smoking crack (70.5%), backloading (i.e., syringe-mediated sharing) (42.9%), receptive syringe sharing (45.1%), and having 2 or more sex partners (54.6%) [$p<0.05$ for all comparisons]. Crossover participants were most likely to inject within the West side drug market area and report shorter stays (median days) at each residence (88.1%, 160.3 days) compared to those who exclusively resided in suburban (56.1%, 310.3 days) and urban (65.5%, 232.2 days) areas. Crossover participants were also significantly ($p<0.05$) more likely than their localized counterparts to report a suburban place of birth, homelessness in the past 6 months, backloading, receptive syringe sharing, sharing syringes with their network, and having $\geq$ 2 sex partners (Table 1). Compared to others, urban participants were significantly ($p<0.05$) more likely to be older and report an urban place of birth and least likely to report injecting with others, receptive syringe sharing, and syringe sharing with their network (Table 1).

Predictors of residential transience, residential movement, and bridging transience

In multivariable logistic regression, controlling for age, race/ethnicity, gender, and education, residential transience was associated with homelessness in the past 6 months, (adjusted odds ratio [aOR]=9.67; 95% CI 3.81-24.5), self-reported HCV positivity (aOR=5.41; 95% CI 1.18-24.7), receptive syringe sharing (aOR=1.74; 95% CI 1.07-2.84), exchanging sex for drugs or

money (aOR=9.30; 95% CI 1.65-52.4), and having any network members that met the definition for crossover in the past year (aOR=3.69; 95% CI 1.15-11.8) (Table 2).

Table 3 reports factors associated with residential movement in multinomial regression and shows that, compared to urban, crossover participants were more likely to report receptive syringe sharing (aOR=2.03; 95% CI 1.02-4.01), 2 or more sex partners (aOR=5.11; 95% CI 1.52-17.3), injecting in the West side drug market area of Chicago (aOR=6.05; 95% CI 1.24-29.5), and more suburban network members (aOR=1.95; 95% CI 1.12-3.38). Individuals with exclusively suburban vs. exclusively urban residence were younger, less likely to report homelessness in the past 6 months, and had more suburban network members.

Among those reporting residential transience (n=96), in a multivariable analysis controlling for age, race/ethnicity, and gender, we examined the association between factors listed on Table 2 and bridging transience. We found no statistical differences between localized and crossover participants for most variables, with the exception that crossover participants were more likely to smoke crack (aOR=3.62; 95% CI 1.16-11.3) and inject within the West side drug market area (aOR=8.78; 95% CI 2.23-34.6) (data not shown).

Geographic analyses

Locations where participants resided, purchased drugs, injected drugs, and met sex partners over the past year by residential movement status are shown in Fig. 1 (a-c). Overall, the median number of residences was significantly ($p<0.05$) higher among crossover (3) compared to suburban (1) and urban (1) PWID (Table 1). As expected, among PWID reporting transience, crossover PWID reported slightly larger median residential movement distances (24.2 miles) compared to suburban (11.1 miles) and urban (5.5 miles) PWID ($p<0.1$)

Most young PWID purchased drugs within the urban West side drug market areas. As expected, most urban PWID (94.6%) reported injecting in urban areas at least once in the past year (Fig. 1a); however, a significant proportion of crossover (88.1%) (Fig. 1c) and a substantial number of suburban PWID (43.2%) (Fig. 1b) also reported injecting in urban areas. Specifically, in the large outdoor West side drug area, crossover PWID were significantly ($p<0.01$) more likely (88.1%) compared to urban (65.5%) and suburban PWID (56.1%) to inject at least once in the past year. Additionally, the proportion of total injection episodes performed in these urban drug markets was also significantly ($p<0.01$) higher among crossover (62.8%) compared to urban (53.4%) and suburban (39.0%) only residents. Crossover (45.3%) and urban (53.4%) PWID were also more likely than suburban PWID (11%) to meet sex partners in urban Chicago ($p<0.01$); crossover PWID were most likely (21%) to meet sex partners within the West side drug market boundaries compared to urban PWID (12% and suburban (2%) ($p=0.06$). Exchange sex was associated with meeting sex partners in Chicago ($p<0.01$) and specifically with meeting sex partners within the West side drug market boundaries ($p<0.01$) (data not shown).

Discussion

For the first time, our study examined patterns of residential instability and associated risk among predominantly non-urban young PWID in the U.S. We identified a sub-group of young PWID (crossover participants) with higher risk behaviors compared to their more residentially localized (urban or suburban) counterparts. These participants may serve as potential 'bridges' between areas. Bridge populations play an important role in the spread of infectious disease by linking subgroups from high and low prevalence settings that would otherwise not have been linked and thus facilitating disease transmission[38,39]. Because they link suburban and urban injection and sexual networks, crossover transients may facilitate transmission of HIV/HCV

from higher (urban) to lower (suburban) HIV/HCV prevalence settings. Identifying bridging populations is important because they represent a potential intervention target, and presence of such bridge populations may explain why increases in HIV and HCV have been observed among suburban PWID populations where prevalence was previously low.

Our study results are consistent with prior research in Midwestern Illinois, U.S. showing that young PWID are predominantly NH-whites from the suburban areas surrounding the city of Chicago (59-82%)[17,19,27,40]. Most young PWID who either reported living exclusively in the suburban (76%) or in both urban and suburban (73%) areas in the past year were born in the suburbs, compared to only 27% of those who reported urban only residence. In Illinois, as with many states, HIV and HCV prevalence in urban[41] areas are dramatically higher than their surrounding suburbs[42]. Centralized urban outdoor drug markets areas were key settings among all PWID for other activities beyond drug purchase, including injecting and meeting injection and sexual network partners. Proximity and interaction with urban areas may be key to understanding the evolution of HCV risk among young suburban PWID. Targeted harm reduction intervention in these areas might have a significant impact on HCV transmission risk over time as individuals age and further advance in their injection career.

Our study results have important implications for interventions, especially with HCV. Recently, the World Health Assembly and World Health Organization (WHO) has recognized the need to prevent and control HCV infection, and proposed that HCV elimination was feasible by 2030 by reducing new chronic infections by 90% and HCV-related mortality by 65%[43]. The recent availability of all oral direct-acting antivirals (DAAs) with high reported cure rates (e.g., >90%[44,45]) that can prevent liver disease progression and HCV transmission,[46,47] combined with prevention and harm reduction strategies, make HCV elimination an attainable goal. However,

due to cost of DAAs, poor linkage to care, low adherence, and possible reinfection[44,48-54], policy development and strategic planning to understand the factors that would most efficiently lead to HCV elimination among PWID are warranted. To address this need, our team has developed mathematical and low-dimensional agent-based models to simulate HCV spread and DAA treatment scale up among PWID in metropolitan Chicago[55,56]. To develop high-dimensional agent-based models that would provide much more realistic predictions of how to design HCV elimination by 2030, the current study's results provide important insights into strategies that might address key subpopulations. In particular, studies of urban PWID have shown that crossover transient individuals can play a central role in the epidemic spread of HCV[24,25]. These individuals often have interactions (residences and sexual relations) both within drug market areas, which are higher risk, and outside of these areas, which is typically lower risk, and thus act as bridges in spreading the disease to new regions (Fig. 1a). Our geographic analyses provide support for potential bridging-related HIV/HCV transmission between areas of higher and lower HIV prevalence. Both urban and crossover PWID performed most injections in Chicago though only approximately half of their reported sex partner meeting locations were in Chicago. While most suburban PWID reported meeting sex partners outside of Chicago, a substantial proportion of injection episodes (43%) among suburban PWID occurred in Chicago, and of these, the majority (83%) were within the West side drug market boundaries. Injection and sexual risk behaviors (e.g., syringe sharing, exchange sex, multiple sex partners) were prevalent in our sample, particularly among crossover PWID, and thus mobility between urban and suburban areas could facilitate the spread of infections by linking networks of varying disease prevalence. Additional research is warranted to understand the mechanisms by which mobility, and crossover transience in particular, may directly impact risk behavior and HIV/HCV transmission.

Residential instability likely leads to vulnerability by increasing the likelihood of injecting in riskier situations (i.e., injecting in public places or in drug markets) or with different and potentially higher risk networks. Prospective studies would help to elucidate the temporal associations between housing instability, changes in network characteristics, and risk. Future research should also focus on the evolution of interaction with drug market areas over the course of a PWID's injecting career. For example, among those who travel long distances to these areas, who are more likely to inject and/or eventually take up residence in these areas and how does relates to their HIV/HCV risk over time? A better understanding of this progression could lead to early identification of risk-prone individuals for targeted intervention.

Limitations

Small sample size limited our power for subgroup analyses and testing of interactions, and resulted in wide confidence intervals for some estimates. The cross-sectional study design did not allow us to determine temporality of associations between individual and network characteristics and transience. The study was limited to a single geographic region (i.e., metropolitan Chicago) and the majority of participants were recruited from a large SEP, which limits the generalizability of the findings to other populations of PWID. Sociodemographic and risk behaviors were only collected for core drug using network members and were not available for those with casual drug using networks. Finally, data were collected via face-to-face interview and drug use behaviors may have been underreported due to social desirability bias. We also used a fairly broad definition of transience (i.e., more than 1 residence in the past 12 months), which may have resulted in diluted effects. Refining the definition of and measurement of residential instability as a construct may help to pinpoint associations with specific risk behaviors and identify mechanisms of impact, as well as facilitating comparison across studies.

Despite these limitations, our study demonstrates high risk behavior associated with residential instability among young suburban PWID, and points to a need for further field research and theoretical modeling to understand mechanisms by which transience impacts HCV and HIV transmission among the new generation of PWID. The association of crossover transience with high risk behavior has important implications for prevention in this population particularly given the potential for disease transmission associated with bridging between areas of high and low HIV/HCV prevalence. Interventions for young PWID may benefit from identifying those who travel between urban and suburban areas, as well as those with residential instability in general, and should address risk associated with transience for this population.

Table 1: Participant characteristics by residential movement status in past 12 months

	Suburban N=60 n (%)	**Crossover N=44 n (%)**	**Urban N= 59 n (%)**	**p-value[a]**
Age				
18-24	25 (41.7)	16 (36.4)	12 (20.3)	0.037
25-30	35 (58.3)	28 (63.6)	47 (79.7)	
Gender				
Male	37 (61.7)	28 (63.6)	40 (69.0)	0.494
Female	23 (38.3)	15 (34.1)	18 (31.0)	
Transgender	0 (0.0)	1 (2.3)	0 (0.0)	
Race/ethnicity				
NH White	45 (75.0)	33 (75.0)	39 (66.1)	0.587
Hispanic	9 (15.0)	7 (15.9)	9 (15.3)	
NH Black/Other	6 (10.0)	4 (9.1)	11 (18.6)	
Educational attainment				
Less than high school	20 (33.3)	9 (20.5)	15 (25.4)	0.278
High school diploma	20 (33.3)	20 (45.5)	30 (50.9)	
Any college	20 (33.3)	15 (34.1)	14 (23.7)	
Place of birth[b]				
Urban Chicago	4 (9.8)	5 (19.2)	22 (59.5)	<0.001
Suburban Chicago	31 (75.6)	19 (73.1)	10 (27.0)	
Out of state	6 (14.6)	2 (7.7)	5 (13.5)	
Regular job	29 (48.3)	21 (47.7)	22 (37.3)	0.411
Homeless in past 6m	19 (32.2)	35 (81.4)	30 (50.9)	<0.001
Self-reported HCV positive	5 (8.3)	7 (15.9)	3 (5.1)	0.164
Years injecting, median (IQR)	5.5 (3-8)	5.5 (3-8.5)	6 (3-9)	0.423
Drug and sexual risk behaviors, past 6m				
Smoked crack	21 (35.0)	31 (70.5)	26 (44.1)	0.001
Poly-injection drug use	18 (30.0)	19 (44.2)	17 (29.3)	0.224
Backloading	13 (21.7)	18 (42.9)	11 (19.0)	0.016
Receptive syringe sharing	18 (30.0)	19 (45.2)	11 (19.0)	0.018
Injected daily	46 (76.7)	33 (75.0)	51 (86.4)	0.272
Injections per day				
1	3 (5.0)	1 (2.3)	2 (3.4)	0.443
2-4	45 (75.0)	27 (61.4)	41 (69.5)	
≥5	12 (20.0)	16 (36.4)	16 (27.1)	
Frequency of injection with others				
Never	6 (10.0)	5 (11.4)	10 (17.0)	0.047
Sometimes	26 (43.3)	30 (68.2)	31 (52.5)	
Often	28 (46.7)	9 (20.5)	18 (30.5)	
Exchanged sex for drugs or money	10 (16.7)	11 (25.0)	9 (15.3)	0.410
Total sex partners				
0	8 (13.3)	5 (11.4)	10 (17.0)	0.033
1	27 (45.0)	15 (34.1)	35 (59.3)	
≥2	25 (41.7)	24 (54.6)	14 (23.7)	

Table 1: Continued				
Geographic variables				
Total residences in the past year, Median (IQR)	1.0 (1.0-2.0)	3 (2.5-4.5)	1.0 (1.0-3.0)	<0.001
Purchased drugs ≥ 1 on West Side	54 (94.7)	39 (95.1)	49 (87.5)	0.899
Purchased drugs in multiple drug market areas	14 (24.6)	10 (24.4)	13 (23.2)	0.984
Inject in multiple locations	19 (33.3)	15 (35.7)	11 (20.0)	0.167
Inject drugs ≥ 1 in West Side drug market area	32 (56.1)	37 (88.1)	36 (65.5)	0.003
Proportion of total injection episodes in West Side drug market area	39.0%	62.8%	50.9%	<0.01
Average stay in days[c], Median (IQR); n	310.3 (184-776.5); 25	160.3 (100.2-301.5); 39	232.2 (153.4-341.5); 28	0.004
Average move distance in miles[c], Median (IQR); n	11.1 (3.1-46.8); 25	24.2 (13.1-64.3); 38	5.4 (2.7-568.0); 28	0.079
Change in distance from drug market[c], Median (IQR)	0 (-3.5, +2.5)	0.24 (-3.1, +5.7)	-0.12 (-2.2, +0.32)	0.718
Network variables				
Core drug using network size, Median (IQR) [n]	3 (2-5) [56]	3.5 (1-7.5) [40]	3 (2-5) [51]	0.599
Mean age of network members, Median (IQR) [n]	27.3 (25.1-30.1) [56]	29.4 (24.3-33.0) [40]	30.7 (27-35.8) [51]	0.003
Proportion of network members homeless, Median (IQR) [n]	0 (0-0) [56]	0 (0-13%) [40]	0 (0-33%) [51]	0.079
Proportion of network members NH white race, Median (IQR) [n]	100% (69-100%) [56]	68% (45-100%) [40]	80% (50-100%) [51]	0.105
Any network member is crossover	11 (19.6)	15 (37.5)	19 (37.3)	0.077
Any network member is suburban	50 (89.3)	19 (47.5)	13 (25.5)	<0.001
Any network member is homeless	7 (12.5)	14 (35.0)	17 (33.3)	0.015
Any network member is HCV positive	12 (21.4)	7 (17.5)	10 (19.6)	0.892
Any network member reports syringe sharing	18 (32.1)	21 (52.5)	11 (21.6)	0.008

a. Pearson Chi-square or Wilcoxon rank-sum test

b. Place of birth was only available for representative sample (n=104; 63%)

c. Among those with multiple residences

Table 2: Results of univariable and multivariable logistic regression: Factors associated with residential transience, N=147

	Univariable OR (95% CI)	p-value	Multivariable OR (95% CI)	p-value
Age 25-30 vs. 18-24	0.72 (0.37-1.42)	0.346	0.95 (0.35-2.62)	0.925
Male gender	0.91 (0.47-1.77)	0.785	3.17 (1.08-9.32)	0.036
NH white race	1.23 (0.62-2.44)	0.556	0.89 (0.32-2.45)	0.820
Education (ref=less than HS)				
High school diploma	1.53 (0.71-3.30)	0.274	1.76 (0.45-6.86)	0.418
Any college	2.96 (1.25-7.01)	0.014	4.91 (1.25-19.3)	0.023
Suburban born[a]	0.99 (0.45-2.17)	0.975	--	--
Regular job	1.11 (0.59-2.08)	0.748	--	--
Homeless in past 6m	6.42 (3.19-12.9)	<0.001	9.67 (3.81-24.5)	<0.001
Self-reported HCV positive	5.09 (1.10-23.5)	0.037	5.41 (1.18-24.7)	0.029
Years injecting	0.99 (0.91-1.07)	0.779	--	--
Drug and sexual risk behaviors, past 6m				
Smoked crack	2.30 (1.21-4.39)	0.011	--	--
Poly-injection drug use	1.99 (0.99-4.01)	0.053	--	--
Backloading	2.76 (1.24-6.14)	0.013	--	--
Receptive syringe sharing	3.25 (1.50-7.01)	0.003	1.74 (1.07-2.84)	0.025
Injected daily	0.59 (0.26-1.35)	0.211	--	--
Injections per day (ref=1)				
2-4	1.15 (0.22-6.00)	0.865	--	--
≥5	2.75 (0.48-15.6)	0.254	--	--
Frequency of injection with others (ref=Never)				
Sometimes	2.37 (0.89-6.33)	0.086	--	--
Often	3.16 (1.11-8.98)	0.030	--	--
Exchanged sex for drugs or money	4.61 (1.66-12.8)	0.003	9.30 (1.65-52.4)	0.011
≥2 sex partners past 6m	3.06 (1.53-6.11)	0.002	--	--
Geographic variables				
Drug markets in multiple places	1.63 (0.75-3.57)	0.218	--	--
Inject in multiple locations	1.88 (0.88-3.98)	0.101	--	--
Inject ≥ 1 in West side drug market area	1.06 (0.53-2.12)	0.870	--	--
Network variables				
Core drug using network size	1.14 (1.01-1.28)	0.039	--	--
Total suburban network members	0.94 (0.82-1.09)	0.434	--	--
Percent network NH white	0.45 (0.15-1.31)	0.141	--	--
Mean age of network members	0.99 (0.94-1.05)	0.761	--	--
Any network member is homeless	4.31 (1.66-11.2)	0.003	--	--
Any network member is crossover	5.02 (2.04-12.3)	<0.001	3.69 (1.15-11.8)	0.028
Any network member reports syringe sharing	2.93 (1.34-6.41)	0.007	--	--
Any network member is HCV positive	2.23 (0.88-5.65)	0.090	--	--

Table 3: Multinomial logistic regression: Factors associated with residential movement (including core drug using network variables), N=147[a]

	Suburban vs. Urban aOR (95% CI)	p-value	Crossover transient vs. Urban aOR (95% CI)	p-value
Age 25-30 vs. 18-24	0.27 (0.09-0.82)	0.021	0.42 (0.13-1.37)	0.151
Male gender	0.86 (0.29-2.57)	0.786	1.43 (0.44-4.64)	0.553
NH white race	2.06 (0.68-6.21)	0.200	1.22 (0.34-4.31)	0.763
Homeless in past 6 months	0.26 (0.08-0.82)	0.022	2.74 (0.79-9.47)	0.111
Receptive syringe sharing past 6m	1.52 (0.84-2.75)	0.171	2.03 (1.02-4.01)	0.043
≥2 sex partners past 6m	2.45 (0.82-7.31)	0.109	5.11 (1.52-17.3)	0.009
Inject drugs ≥ 1 in West Side drug market area	0.63 (0.22-1.80)	0.387	6.05 (1.24-29.5)	0.026
Total suburban network members	2.53 (1.51-4.24)	<0.001	1.95 (1.12-3.38)	0.018

aOR=adjusted odds ratio. Estimates are adjusted for other variables presented.
Models exclude 1 transgender individual.
a. 17 observations were excluded due to missing data

FIG. 1 CAPTION:

Fig 1a-c. Places where young (a.) urban (n=59), (b.) suburban (n=60), and (c.) crossover (n=44) PWID resided, purchased drugs, injected drugs, and met sexual partners in the past year

Urban was defined as having ≥ 1 residence(s) exclusively within the Chicago city limits; suburban was defined as having ≥ 1 residence(s) exclusively in the suburban areas surrounding Chicago; and crossover was defined as having ≥1 residence(s) in urban *and* ≥1 residence(s) in suburban areas in the past year. Maps include multiple locations per person.

FIG. 1A-C

Fig.1a.

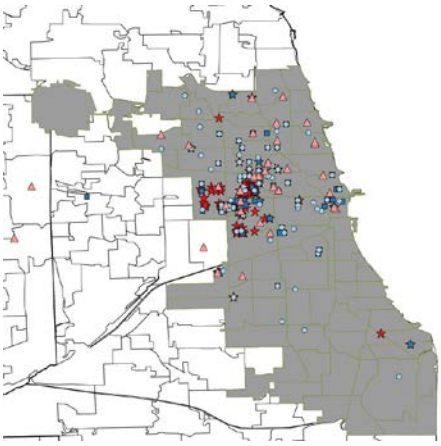


Fig.1b.

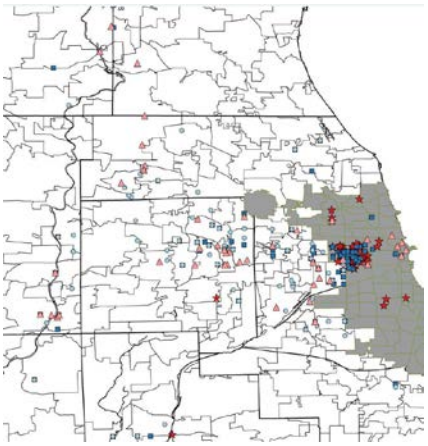
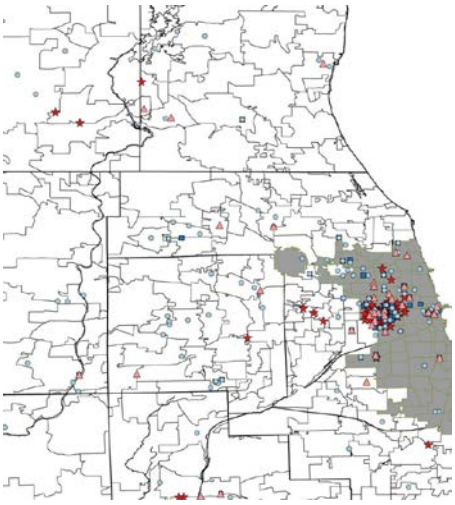


Fig. 1c.



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